

ABDURREHMAN SHINWARI

Portfolio

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PROFESSIONAL SUMMARY

Computer Engineer with hands-on experience building machine-learning and deep-learning applications using Python, scikit-learn, and TensorFlow/Keras. Developed an end-to-end AI healthcare application combining symptom-based disease prediction, medical-image classification, medicine recommendations, and Flask web integration. Interested in AI/ML engineering, model deployment, web development and user-facing intelligent applications.

TECHNICAL SKILLS

Programming: Python, Next.js, C++

Machine Learning: Random Forest, Naive Bayes, K-Nearest Neighbors, classification, data preprocessing, model evaluation

Deep Learning: CNNs, transfer learning, MobileNetV2, EfficientNetB0

Libraries & Frameworks: TensorFlow/Keras, scikit-learn, pandas, NumPy, Flask

Web & Tools: HTML, CSS, JavaScript, Google Colab, Visual Studio Code

Image Processing: Image resizing and normalization, feature extraction, GLCM

PROJECTS

AI Healthcare Assistant | *Python, Flask, TensorFlow/Keras, scikit-learn, HTML, CSS, JavaScript*

- Led technical development of a working multimodal healthcare decision-support application integrating symptoms, skin-image, and chest X-ray analysis.
- Built a Random Forest pipeline to classify 10 diseases from textual symptoms and provide follow-up questions, precautions, medicine recommendations, confidence scores, and a downloadable PDF medical report.
- Integrated MobileNetV2 transfer learning to classify five skin conditions: acne, chickenpox, impetigo, psoriasis, and tinea (ringworm).
- Integrated EfficientNetB0 to classify chest X-rays as normal, pneumonia, or tuberculosis, and connected all models to a Flask web interface.
- Independently implemented the core technical functionality and explained the system to team members for project presentations.
- GitHub Repository: github.com/Abdur-Rehman-170/ai-healthcare-assistant

Spam Email Classifier | *Python, scikit-learn, pandas, Naive Bayes, Google Colab*

- Developed a Naive Bayes text-classification model to categorize email messages as spam or legitimate.
- Prepared the email dataset and implemented the model-training and evaluation workflow in Google Colab.

Hospital Management System | *C++, Object-Oriented Programming*

- Developed a console-based application for managing patient records and hospital appointments.
- Applied C++ classes, inheritance, and vectors to structure application data and functionality.

Student Pass/Fail Prediction | *Python, scikit-learn, Random Forest, Google Colab*

- Developed a Random Forest classifier to predict whether a student would pass or fail using previous quiz scores, study hours, and attendance.
- Prepared the dataset and implemented the classification workflow in Google Colab.

EDUCATION

Bachelor of Science in Computer Engineering | September 2022 - August 2026

COMSATS University Islamabad, Abbottabad Campus | Abbottabad, Pakistan